



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE • INDIA

A Project Report on

TITTLE OF PROJECT

Submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

Name of Specialization

by

Name 1 RegNo 1

Name 2 RegNo 2

Name 3 RegNo 3

Name 4 RegNo 4

Name 5 RegNo 5

Under the Guidance of

Guide Name

and

Co-Guide Name

Department Name

School of Engineering and Technology,

CHRIST (Deemed to be University),

Kumbalagodu, Bengaluru - 560 074

March-2025



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE · INDIA

Vision

“Excellence and Service”

Mission

“CHRIST (Deemed to be University) is a nurturing ground for an individual’s holistic development to make effective contribution to the society in a dynamic environment.”

Core Values

Faith in God

Moral Uprightness

Love of Fellow Beings

Social Responsibility

Pursuit of Excellence



CHRIST
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Department Vision

“Department Vision”

Department Mission

M1: Department Mission1.

M2: Department Mission2.

M3: Department Mission3.

M4: Department Mission4.

M5: Department Mission5.

Program Educational Objectives (PEOs)

PEO1: PEO1

PEO2: PEO2

PEO3: PEO3

PEO4: PEO4

PEO5: PEO5

Program Outcomes (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1: PSO1

PSO2: PSO2

PSO3: PSO3

PSO4: PSO4

PSO5: PSO5



School of Engineering and Technology

Department Name

CERTIFICATE

This is to certify that **Name 2** has successfully completed the project work entitled “**Tittle of Project**” in partial fulfillment for the award of **Bachelor of Technology** in **Name of Specialization** during the year **2024-2025**.

Guide Name

Guide Designation

Co-Guide Name

Co-Guide Designation

Name of the HOD

Head of the Department

Dr Raghunandan Kumar R.

Dean



School of Engineering and Technology

Department Name

BONAFIDE CERTIFICATE

It is to certify that this project titled “**Tittle of Project**” is the bonafide work of

Name	Reg. No.	Department
Name 1	RegNo 1	Electrical & Electronics Engineering
Name 2	RegNo 2	Mechanical Engineering
Name 3	RegNo 3	Mechanical Engineering
Name 4	RegNo 4	Electrical and Electronics Engineering
Name 5	RegNo 5	Mechanical Engineering

Examiners [Name and Signature]

1.

2.

Name of the Candidate :

Register Number :

Date of Examination :

Industry Certificate

**Certificate PDF File has to kept in the Pictures Folder with the name
IndustryCertificate.pdf**

Acknowledgement

We would like to thank **Dr Rev Fr Joseph C. C.**, Vice Chancellor, CHRIST (Deemed to be University), **Dr Rev Fr Viju P. D.**, Pro Vice Chancellor, **Dr Fr Sony J. Chundattu**, Director, School of Engineering and Technology, **Fr Shijin P. J.**, Assistant Director, School of Engineering and Technology, CHRIST (Deemed to be University), **Dr Raghunandan Kumar R.**, Dean, and **Dr Mary Anitha E. A.**, Associate Dean, for their kind patronage.

We would also like to express sincere gratitude and appreciation to **Name of the HOD**, Head of the Department, Department Name for giving me this opportunity to take up this project.

We also extremely grateful to my guide, **Guide Name**, who has supported and helped to carry out the project. His constant monitoring and encouragement helped me keep up to the project schedule.

We also extremely grateful to my co-guide, **Co-Guide Name**, who has supported and helped to carry out the project. His constant monitoring and encouragement helped me keep up to the project schedule.

If outside the college-mention the organisation and the concerned people, like head of the organisation, guide and any other person you want to thank. All faculty and non-teaching staff. You may acknowledge your parents or any who supported you.

Declaration

We, hereby declare that the project titled “**Tittle of Project**” is a record of original project work undertaken for the award of the degree of **Bachelor of Technology** in **Department Name**. We have completed this study under the supervision of **Guide Name**, Guide Department and **Co-Guide Name**, Co-Guide Department.

We also declare that this project report has not been submitted for the award of any degree, diploma, associate ship, fellowship or other title anywhere else. It has not been sent for any publication or presentation purpose.

We have executed this project with code of research conduct as prescribed by the university.

Place: School of Engineering and Technology,
CHRIST (Deemed to be University),
Bengaluru

Date: 15-03-2025

Name	Register Number	Signature
Name 1	RegNo 1	
Name 2	RegNo 2	
Name 3	RegNo 3	
Name 4	RegNo 4	
Name 5	RegNo 5	

Abstract

All reports or thesis, both BTech and MTech, must have an abstract included in it. An abstract is a summary of what is in the report or thesis.

To Add/Edit abstract page open /Primitive/Abstract.tex file

Keywords: Word1, Word2

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GLOSSARY

Item	Description
Atomic Weight	The atomic weight is the average of the isotope weights weighted for the isotope distribution and expressed on the ^{12}C scale
Ohms Law	Voltage Prop to current
IC Engine	I nternal C ombustion E ngine
AIR	A ll I ndia R adio
PSO	P article S warm O ptimization
CT	C urrent T ransformer
RDBMS	R elational D ata B ase M anagement S ystem
DSM	D emand S ide M anagement
PLC	P eak L oad C ontroller
GUI	G raphical U ser I nterface
Speed of Light, c	$2.997\,924\,58 \times 10^8 \text{ ms}^{-\text{s}}$
π	$= 3.14$
P	Power in W (Js^{-1})
ω	Angular frequency in rads^{-1}

Chapter 1

INTRODUCTION

The report shall not have more than five chapters. The title or heading of these chapters are also fixed except for the Chapter named ‘Actual work’. Students have the flexibility to choose the title for this chapter and its sub chapters.

Introduction chapter for both BTech and MTech. The chapter shall also have the following four mandatory sub-chapters:

1.1 Context and Background Study

Introduce the topic and provide relevant background information. Explain why the topic is important and why it warrants further research. Briefly review the current state of knowledge in the field

1.2 Need Analysis

Systematic process of identifying the needs, problems, or gaps between a current state and a desired state. Explain the disparities between the existing conditions and the desired state in a particular project.

1.3 Problem Identification

The identified problem shall be formulated in a systematic manner and provide clarity to decide on the problem statement and objectives

1.4 Problem Formulation

Under this the reason for choosing the particular problem or title for the project shall be explained along with the thought process that was involved in doing so.

Chapter 2

RESEARCH METHODOLOGY

MTech project report or thesis must have this chapter. This is an optional chapter for BTech, but highly recommended. Along with a brief introduction to research methodology and its fundamentals, under this chapter, it is required to include the methodology adopted for entire research process of the project work, including the preliminary research or background study carried out to identify and formulate the problem. Research methodology is the most important aspect of any research based project work. All Christ University libraries have books and other literatures on research methodology. There are a lot of MOOCS courses too on research methodology and completing one is highly recommended for all MTech students if they do not have it as one of their courses.

Chapter 3

LITERATURE SURVEY AND OBJECTIVES

Any research based project is incomplete without a literature survey and review. Hence, this chapter is mandatory to both MTech and BTech projects. The following sections and flow should be maintained.

3.1 Critical Review of Literature

Lirerature survey of domains related to the research or project.

3.2 Literature Collection & Segregation

Classification, analysis and comprehensive understanding of topic.

3.3 Summary of Literature

Overall summary of the lirerature review.

3.4 Identification of Gaps

Based on the literature review, enumerate the research or market gap.

3.5 Need for Research/Project

Explain the need and significance of the project.

3.6 Problem Statement

Write problem statement based on the research gap and need analysis.

3.7 Objectives

List of project objectives along with the problem statement shall be provided in this section.

1. Objective 1
2. Objective 2
3. Objective 3
4. Objective 4
5. Objective 5

3.8 Limitations of the Study

This is an academic research based project and has its own limitations with respect to time and other constraints. Also, there might be other limitations of the project work depicted in the reported that may not be obvious from the title. It is very important to mention those limitations and avoid any misconceptions in the minds of the readers and/or evaluators.

Chapter 4

ACTUAL WORK

Upon completion of identifying & formulating the research problem, and carrying out the necessary literature survey and review, the actual work on the project is taken-up. This chapter is dedicated to the actual work done by students. Hence, the chapter name and sub-chapter names are not fixed. It is left to the discretion of the students with appropriate guidance from their respective supervisors. However, one or more of the following aspects (as applicable) shall be covered in this chapter:

- Methodology of the study or actual work (different from research methodology)
- Experimental and/or analytical work completed in the project
- Modeling, Analysis and Design
- Prototype and testing

4.1 Methodology for the Study

4.2 Experimental Work

4.3 Analytical Work

4.4 Modeling, Analysis & Design

4.5 Prototype & Testing

4.6 Algorithms and Flowcharts

4.7 Codes / Standards

Sample LaTeX Typesetting

Figure Vector Graphics EPS Format [Figure 4.1]

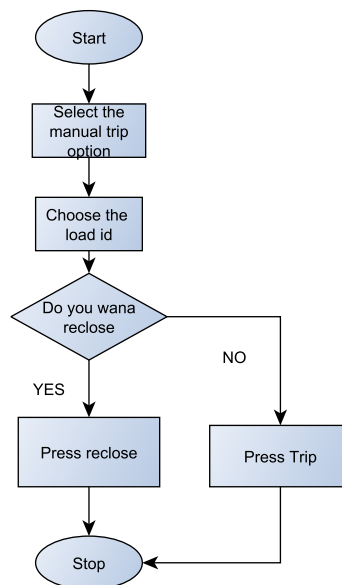


FIGURE 4.1: Flowchart of Manual Controller

Figure JPEG/JPG Format [Figure 4.2]

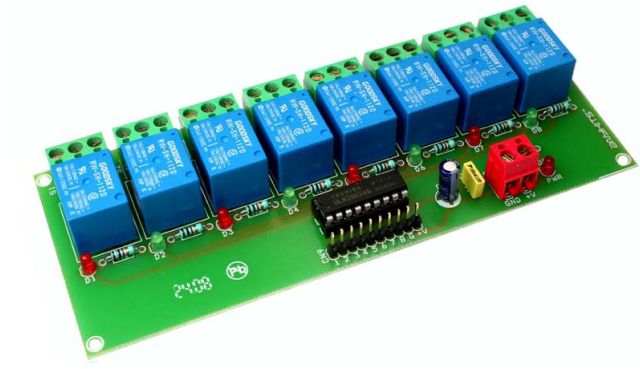


FIGURE 4.2: Relay Board

Table Refer [Table 4.1]

TABLE 4.1: Student Marks

Name	Marks
Ajay	10
Vinay	20

Cross References: Citation, Index, Reference, Equation reference This is the methodology for the entire project work which includes even the process of deciding on the project title, objectives , etc. This is mandatory for MTech and optional for BTech)[2]. The data is shown in Table 4.1. The equation shown in Equation (4.1)

Inline Equation This is my equation. $f = ma \pm \alpha \Delta \begin{bmatrix} 1 & \chi \\ -1 & 0 \end{bmatrix}$, which is appearing in between some text.

Equation without Numbering

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Equation with Numbering

$$\dot{X} = \begin{bmatrix} 1 & p \\ 2 & \alpha \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + Bu \quad (4.1)$$

Algorithm 1 Addition of two 8 bit numbers

- 1: Start
 - 2: Input a and b
 - 3: c=a+b
 - 4: Output c
 - 5: stop
-

Algorithm Format

Enumeration Format The following are the different flavor of Tex systems

1. TeXLive TeX System
 - (a) TeXLive for Windows
 - i. TeXLive
 - ii. ProTex
 - (b) MacTeX for Mac
2. MikTeX TeX System

Bullets Format The following are the advantages of LaTeX,

- L^AT_EX is highly portable and free.
 - Contribute to TUG
 - Promote Free Softwares
- Operating-system independent.
- Complex scientific documents can be created automatically.
- High quality math typesetting.

Program Inclusion Program file present in other directory can be embedded into the report.

```
; Addition of two 8 bit numbers
ORG 0000H ; Starting address of the program
MOV R1,#10H ; First number
MOV R2,#20H ; Second number
MOV A,R1 ; Input first number from R1 to A
MOV B,R2 ; Input second number from R2 to B
ADD A,B ; Add A and B
MOV R3,A ; Store the result in R3
RET
```

Verbatim Text Include text as it is.

The additional database schema is shown below which is used to store all the configuration and transaction data.

```
CREATE TABLE 'controller_config' (
  'load_id' int(11) NOT NULL,
  'ct_constant' double DEFAULT NULL,
  'pt_constant' double DEFAULT NULL,
  'samples' int(11) DEFAULT NULL,
  'delay' double DEFAULT NULL,
  PRIMARY KEY ('load_id')
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
SELECT * FROM loadcontroller.load_details;
```

Chapter 5

RESULTS, DISCUSSIONS AND CONCLUSIONS

Here, the results of the project work (literature survey and review along with actual work) shall be listed and discussed in detail with appropriate arguments (result analysis) leading to logical conclusions. The list of conclusions should sync with the project objectives. The scope for future research and development in the field of the current project work must also be included in this chapter.

All the sections are mandatory for all projects.

5.1 Results & Analysis

5.2 Comparative Study

Content

5.3 Discussions

Content

5.4 Project Cost Estimation

Provide project execution cost and product/service cost estimation in the form of Table.

TABLE 5.1: Project Execution Cost

Project Execution Cost					
No	Category	Description	Unit Cost (₹)	Quantity	Cost
1	Material Costs	Raw materials (electronics, electrical, wood, steel, etc.)	1000	2	2000
2	Equipment Costs	Machinery and tools (rent or purchase)	100	2	200
3	Labor Costs	Wages of outsourced work	5000	1	5000
4	Transportation	Shipping and delivery of materials or equipment	250	4	1000
Total Cost					8200

TABLE 5.2: Product/Service Cost Estimation

Product/Service Cost Estimation					
No	Item	Description	Unit Cost (₹)	Quantity	Cost
1	Item1	Description1	1000	2	2000
2	Item2	Description2	100	2	200
3	Item3	Description3	5000	1	5000
4	Item4	Description4	250	4	1000
Total Cost					8200

5.5 Project Impact

Provide detailed explanation on the impact of professional engineering solutions in societal and environmental contexts.

5.5.1 Environmental Impact

5.5.2 Societal Impact

5.5.3 Economical Impact

5.6 SDG Goal Compliance

Provide detailed explanation on how your project addressing SDG goal/s.

Justification and the relevant SDG Goal

Impact study and reduction of carbon footprint, if any

5.7 Conclusions

Conclusion should be on new page and the same should come here.

Conclusions on stated objectives (Conclusions to be numbered correlating with stated objectives and not bulleted)

Conclusions beyond stated objectives

General conclusions, if any

5.8 Scope for Future Work

Future scope should be on new page and the same should come here.

Indicate broad objectives

Suggested methodology/s

Bibliography

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PUBLICATION DETAILS

1. S.K. Kenue and J.F. Greenleaf, “Limited angle multifrequency diffraction tomography,” *IEEE Trans. Sonics Ultrason.*, vol. SU-29, no. 6, pp. 213-217, July 1982.

Appendix A

Appendix A Title

Since the chapters are numerically numbered, the appendices shall be numbered using alphabets (English capital letters). The items that can be inserted as appendices are (list is not exhaustive):

- Project synopsis or proposal (if submitted before starting the project)
- Photos
- Software model analysis reports (these shall not be inserted in the main body of the report)
- Project schedules
- Selected material from the data collected
- Miscellaneous analysis and reports

A.1 Appendix A Section 1

A.1.1 Appendix A Subsection for Section 1

A.2 Appendix A Section 2

Appendix B

Hardware and Software Tools

B.1 Appendix B Section 1

B.2 Appendix B Section 2

B.3 Appendix B Section 3

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